

# Rui Zeng

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🏢 Shenzhen Technology University · 🎓 Computer Science and Technology

**Fields of Interest:** Robotics, Embodied AI, Reinforcement Learning, Sim-to-Real, Human-Robot Interaction

## 🎓 Education

Sep. 2023 | **Shenzhen Technology University** · School of Artificial Intelligence · Computer Science and Technology · B.Eng.  
Present

## 🔧 Technical Skills

- › **Languages:** Python, C/C++
- › **Robotics & AI:** ROS/ROS2, MoveIt, Isaac Gym / Isaac Sim, PyTorch, SLAM (Cartographer, FAST-LIO)
- › **Embedded & Hardware:** ESP32, STM32, RDK X3/X5, Motor Control, Sensor Integration
- › **Tools & Platforms:** Git, Docker, Linux, MuJoCo

## 📄 Publications

- › S. Luo\*, **R. Zeng\***, J. Lyu, Q. Li. "ATOM: A Training Framework for Humanoid Behavioral Foundation Models under Imperfect Data." Submitted to **NeurIPS 2026**. (\*Equal contribution, under review.)

## 🔬 Research Experience

### Humanoid Whole-Body Control

Mar. 2026 – Present

- › Developed an end-to-end humanoid motion tracking pipeline integrating skeletal motion extraction, retargeting, RL-based policy training (AMP), simulation validation, and physical deployment.
- › Investigated robust humanoid motion tracking under imperfect data, building demonstrations connecting human motion input, simulation tracking, and Sim2Real transfer.
- › Work resulted in a co-first-author paper submitted to **NeurIPS 2026** (see Publications).

### Morphology-Aware Quadrupedal Locomotion · HKUST(GZ)

Feb. 2026 – May 2026

- › Collaborated with senior researchers at HKUST(GZ) on reinforcement learning-based quadrupedal locomotion under embodiment uncertainty.
- › Focused on morphology adaptation through per-limb randomization, target-height conditioning, proprioceptive leg-length inference, and robustness evaluation across A1 simulations and Unitree Go2 hardware.

## 🔗 Project Experience

### ICRA 2024 Sim2Real Challenge

2024

- › Developed a perception-and-control pipeline for a mobile manipulator, combining SLAM navigation, object detection, grasp pose prediction, and robotic manipulation.
- › Implemented localization, autonomous navigation, robotic arm inverse kinematics, and module integration; transferred the full stack from simulation to a physical robot.
- › Won **First Prize / 2nd Place** in the ICRA 2024 AgileX Sim2Real Challenge.

## STEM Education Robot Kit · Co-Founder

Dec. 2025 – May 2026

- › Identified an unmet market need through interviews with STEM educators: existing products failed to integrate hardware assembly with real software programming; adopted ROS as the middleware to bridge low-level hardware control and high-level software education.
- › Designed a robot kit paired with a custom web visual programming platform to lower the learning barrier for students while preserving exposure to real robotics frameworks.
- › Drove the product from prototype to production: built early prototypes, collected educator feedback, validated at industry expos, and iterated on usability and curriculum fit.
- › Delivered a commercially viable product now **deployed to institutions in multiple cities nationwide**, completing the full journey from pain point discovery to market deployment.

## Cathay Pacific Hackathon 2025 – Captain Milo

Nov. 2025

- › Attended industry masterclasses at **Google, Microsoft, Alibaba, and HAECO** on AI/ML, aviation operations, and product innovation.
- › Served as Tech Lead for a 5-person team; built an edge-AI companion device for unaccompanied children on flights, featuring a hybrid cloud-edge architecture on RDK X5.
- › Designed the LLM interaction pipeline with multilingual ASR/TTS (Cantonese, English, Mandarin), emotion-adaptive dialogue, real-time parental alerts via MQTT, and a crew monitoring dashboard.
- › Delivered a working live demo with 3D-printed prototype in a 48-hour sprint; **Finalist (Top 20 / 100+ teams)**.

## RoboMaster Engineering Robot

2024 – 2025

- › Developed a robotic manipulator for the university RoboMaster engineering robot, implementing inverse kinematics and end-effector control through ROS and MoveIt.
- › Designed a custom teleoperation controller for intuitive manipulation during competition tasks.
- › Competed in **RoboMaster University Championship** (founded by **DJI**), a large-scale multi-robot confrontation competition among 400+ university teams.

## Awards

|                                                           |              |           |
|-----------------------------------------------------------|--------------|-----------|
| ICRA 2024 AgileX Sim2Real Challenge                       | First Prize  | May 2024  |
| ICRA 2025 Sim2Real Challenge                              | Second Prize | Jun. 2025 |
| RoboMaster University Sim2Real Challenge                  | Second Prize | May 2024  |
| 2025 RoboMaster University Championship                   | Third Prize  | Aug. 2025 |
| 2025 RoboMaster University Championship Engineering Robot | Third Prize  | Aug. 2025 |
| China Computer Federation 2024 Tech Marathon              | Third Prize  | Oct. 2024 |
| RAICOM Treasure Hunt                                      | Third Prize  | Aug. 2024 |